BlueFairy Forms

Installation & Features Guide

Zero-asset-bloat WordPress forms with Anti-Spam integration,
GDPR-safe storage, rate limiting, and no jQuery.

Version 1.1.10

FREE

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Requirements & Installation

Requirement	Minimum
PHP	8.1+
WordPress	6.4+
BlueFairy Anti-Spam	Optional (strongly recommended)

1. Upload `bluefairy-forms/` to `wp-content/plugins/`
2. Activate via **Plugins > Installed Plugins**
3. On first activation a default “Contact” form is created (ID 1 on a fresh install)
4. Place `[bluefairy_form id="1"]` on any page, using the ID shown in **Tools > BlueFairy Forms** — done

Quick Start

Creating a form

Forms are listed at **Tools > BlueFairy Forms**, with each form’s ID and shortcode. **Add New** asks for a title, a slug (lower-case letters, digits and hyphens) and the form configuration as JSON; see the Form Fields Reference below.

Shortcode

```
[bluefairy_form id="1"]
```

Replace `1` with the form ID shown in **Tools > BlueFairy Forms**.

Gutenberg Block

In the block editor, add a new block and search for **BlueFairy Form**. Select it and enter your form ID under **Form Settings** in the block sidebar.

Form Fields Reference

Field type	JSON `type`	Notes
Single-line text	`text`	Supports `minlength`, `maxlength`, `pattern`

Email	`email`	Must be a valid e-mail address. Give it the id `email` (see below).
Telephone	`tel`	Digits, spaces, `+`, `-` and brackets; 3–30 characters
URL	`url`	Must be a valid URL
Number	`number`	Must be numeric
Date	`date`	`YYYY-MM-DD`
Textarea	`textarea`	Supports `minlength`, `maxlength`
Dropdown	`select`	Takes an `options` array; the value must be one of them
Radio buttons	`radio`	Takes an `options` array; the value must be one of them
Checkbox	`checkbox`	Single checkbox, value = 1

Every field takes `id`, `type` and `label`, and optionally `required`, `placeholder`, `pattern`, `minlength`, `maxlength` and `options`. All validation runs on the server; when a field fails, the visitor is returned to the form with a message naming it.

The Email column, the hashed e-mail used by the privacy tools, and the GDPR export and erase all come from the field whose id is `email`. Use that id for the visitor's address.

Form-level keys: `submit_label`, `success_message`, `consent_text` (adds a consent checkbox that must be ticked: a submission without it is refused with "Please tick the consent box to send this form." and nothing is stored; the text and the time are stored with the submission) and `redirect_url` (must be on your own site; any other address falls back to the success message).

Example config JSON

```
{
  "schema_version": "1.0",
  "submit_label": "Send Message",
  "success_message": "Thank you! Your message has been sent.",
  "consent_text": "I agree to the privacy policy.",
  "redirect_url": "/thank-you/",
  "fields": [
    { "id": "name",    "type": "text",     "label": "Your Name",     "required": true },
    { "id": "email",   "type": "email",    "label": "Email Address", "required": true },
    { "id": "message", "type": "textarea", "label": "Message",       "required": true }
  ]
}
```

Anti-Spam Integration

When **BlueFairy Anti-Spam** is active, BlueFairy Forms delegates all spam checking to it via two hooks:

- `do_action('bluefairy_antispam_render_fields', $form_type)` — renders honeypot and time-trap hidden fields
- `apply_filters('bluefairy_antispam_check', false, $post_data, $form_type, $ip, $email)` — returns true if spam

Form type: `bffrm_{form_id}` (e.g. `bffrm_1`). While Anti-Spam is active the built-in honeypot and time trap below are not used; rate limiting still applies.

While Anti-Spam is not active, administrators see a dismissible notice recommending it in the WordPress admin. Dismissing it hides it for 30 days.

Spam Protection Fallback

Without Anti-Spam, a self-contained honeypot and time-trap protects every form:

- **Honeypot:** one hidden field per form, its name chosen from a pool of eight realistic names; a submission that fills it is blocked. It can be switched off in Settings.
- **Time trap:** an HMAC-SHA256-signed token records when the page was loaded. A submission is blocked when it arrives faster than the Minimum Submission Time (default 3 seconds), more than 24 hours after the page loaded, or with the token missing or altered.

Logged-in users skip both checks.

Rate Limiting

Each form enforces a per-IP rate limit via atomic `INSERT … ON DUPLICATE KEY UPDATE` on `bffrm_rate_limits`. Window: one clock minute. Default max: 5 per IP per form per minute (configurable from 1 to 60). IP addresses are stored only as an HMAC hash.

The address counted is the connection's own (`REMOTE_ADDR`). Forwarding headers are believed only from a proxy you trust, because anyone can send them: before 1.1.9 the first `CF-Connecting-IP`, `X-Forwarded-For` or `X-Real-IP` was believed, so a sender with a new header on every request was never limited.

- **Cloudflare** works with no setup: when the connection comes from one of Cloudflare's published address ranges (built into the plugin), `CF-Connecting-IP` is used. The same header from any other address is ignored. Filter `bffrm_cloudflare_ranges` changes the list.

- **Another proxy or load balancer**: list its addresses or CIDR ranges under **Trusted Proxies** in Settings (or filter `bffrm_trusted_proxies`). From those addresses only, `X-Forwarded-For` is read right to left and the first address that is not itself a trusted proxy is used, then `X-Real-IP`.

Page Caches

A page that shows a form is never served from a page cache: BlueFairy Forms marks it uncacheable (`DONOTCACHEPAGE`, honoured by W3 Total Cache, WP Super Cache and LiteSpeed Cache, and no-cache headers for browsers and proxies). Should a cache serve it anyway — a CDN, or a cache that ignores the constant — the form fetches fresh security tokens when it loads, so it still works. A form left open for more than a day asks the visitor to send it again; it is never thrown away as spam.

Every refused submission is listed under **Refused submissions** at the foot of **Tools > BF Forms — Submissions**: when, which form, and why (spam, too fast, page open too long, too many submissions, a field that did not validate, and so on). Nothing the visitor typed is kept. The list follows the audit log's retention, 90 days by default.

Submissions Log

Tools > BF Forms — Submissions lists submissions newest first, 50 per page. **View Submissions** on a form in **Tools > BlueFairy Forms** shows that form only.

Column	Notes
ID	Auto-increment
Form	Form title
Submitted At	UTC `datetime(6)`
Email	The value of the `email` field, if provided; a keyed hash of it is stored alongside for the privacy tools
Status	`new`, or `erased` after a personal-data erasure
View	Every submitted field, plus the consent text and time when consent was given

Admin Email Notifications

On every new submission, a plain-text email is sent with the form name, submission ID, all field values, and a direct admin link. It goes to the Notification Email in Settings, or the site admin email when that is blank. Its Reply-To is the visitor (the `name` and `email` fields; none without a valid address), so replying reaches them; filter `bffrm_notification_headers` changes the headers.

GDPR Tools

- **Exporter** — exports all submissions matching a user's email
- **Eraser** — anonymises matching submissions: the email, its hash, the IP and user-agent hashes, the field values and the consent text are cleared and the status becomes `erased`. The row itself is kept.

Both run from WordPress's own **Tools > Export Personal Data** and **Tools > Erase Personal Data** screens.

Audit Log

Events are written to the `bffrm_audit_log` database table; there is no admin screen for it. Rows older than 90 days are pruned by a daily scheduled task.

Action	When
`submission.created`	New submission saved
`form.created` / `form.updated` / `form.deleted`	Admin form CRUD

WP-CLI Reference

```
wp bluefairy-forms list-forms [--format=<table|json|csv|ids>]
wp bluefairy-forms doctor
```

`list-forms` lists every form (ID, title, slug, status, created). `doctor` checks that all four tables (`bffrm_forms`, `bffrm_submissions`, `bffrm_rate_limits`, `bffrm_audit_log`) exist, that the hashing key and the settings are stored, reports whether BlueFairy Anti-Spam is active, and counts the forms.

Settings Reference

Tools > BF Forms — Settings.

Setting	Default	Description
Enable Forms	On	Off: every form refuses submissions with “This form is not accepting submissions at the moment.” and nothing is stored
Notification Email	Site admin email	Where new-submission alerts are sent
Rate Limit (per minute)	5	Max submissions per IP per minute per form (1–60)
Minimum Submission Time	3 seconds	Faster submissions are blocked by the time trap (1–30)
Built-in Honeypot	On	Used only while BlueFairy Anti-Spam is not active
Trusted Proxies	Empty	Addresses or CIDR ranges (one per line) of your own proxies or load balancers; see Rate Limiting. Anything else is dropped when saved. Not needed for Cloudflare.
Remove Data on Uninstall	Off	When on, deleting the plugin drops its four tables and removes its options. Cannot be undone.

Multisite

When a new site is added to the network, all four tables are created for it automatically on `wp_initialize_site`. Each subsite uses its own table prefix. Deleting the plugin on a network cleans every site that has **Remove Data on Uninstall** on (before 1.1.9 only the main site was cleaned).

Frequently Asked Questions

Does it load jQuery or external scripts?

No. The form is plain HTML with one small stylesheet, which loads only on pages that show a form. The plugin adds no front-end JavaScript.

Can I redirect to a thank-you page?

Yes — add `"redirect_url": "/thank-you/"` to the form config JSON. The page must be on your own site.

What you are missing

Everything in this guide is the free plugin. **BlueFairy Forms Pro** is where the plugin stops being a contact form and starts being infrastructure — encrypted submissions, webhooks into your own systems, CSV export, file uploads, and CAPTCHA.

Feature	Pro	Free
Encryption at rest (libsodium, with key rotation)	Yes	—
Webhooks with HMAC signature and retry	Yes	—
Streamed CSV export	Yes	—
File uploads with MIME validation	Yes	—
CAPTCHA (reCAPTCHA v3, hCaptcha, Turnstile)	Yes	—
HTML confirmation emails to the submitter	Yes	—
Custom From header and multiple recipients	Yes	—
Form scheduling and submission limits	Yes	—
Dashboard submission stats widget	Yes	—
Forms, Gutenberg block and shortcode	Yes	Yes
Spam protection and rate limiting	Yes	Yes
GDPR-safe storage, export and erasure	Yes	Yes

If you are storing anything a person would mind losing — enquiries, applications, anything with a name and an email attached — it is sitting in your database in plain text right now. Pro encrypts it, and gives you the export and integration tools to do something useful with it.

Get BlueFairy Forms Pro — wordpress.bluefairydevelopment.com/product/bluefairy-forms-pro/
Installs alongside this plugin. Keep BlueFairy Forms active and your existing forms, submissions and settings carry straight over — nothing to migrate.

License & Support

GPL-2.0-or-later. Free.

Support and bug reports: sign in or create an account at the shop and open a support ticket at wordpress.bluefairydevelopment.com/my-account/support/.